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The crisis of quantum gravity

Is radical action needed?



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 cite

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The fact that a theory of quantum gravity hasn't been found reflects a crisis in physics. If we have a clearer understanding of the question "what is quantum gravity", we will be better equipped to find our answer, writes Karen Crowther.

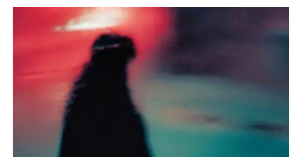
We are all familiar with gravity. We observe its effects in the drooping of our bodies, the falling of sand through an hourglass, and the passing of the years as the Earth orbits the Sun. Our best description of gravity is Einstein's theory of general relativity, which tells us that the effects we observe are due to the interaction of objects with spacetime. While this theory is remarkably successful at describing the universe, physicists have long wanted to do away with it. For more than 70 years, they have been trying to find its successor: a theory of quantum gravity.

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The fact that a theory of quantum gravity hasn't been found reflects a crisis in physics. But it is not obvious what the crisis is. How might we "diagnose" it? Here, I explore this question, and suggest that the answer is connected to another question—what is quantum gravity?

The reason that physicists want a theory of quantum gravity is not that there are any anomalous observations or unexplained experimental results that indicate a need to replace general relativity. Indeed, until recently, there were no observational results agreed to be evidence of quantum gravitational effects, and none of the attempts at a theory made any testable predictions. Although the empirical situation is no longer so bleak, the data available is still unlikely to be sufficient to lead to a new theory or strongly support any of the current attempts at finding one. And now, some physicists are starting to dissent.

Dissenting physicists argue that there are flaws in the formalism and methodology of some of the approaches to quantum gravity, particularly string theory. The underlying causes of the crisis, according to these physicists, range from psychosocial factors such as academic groupthink on the one hand to philosophical and scientific factors on the other.

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What is the Problem? We Don't Know the Problem

Here, I would like to present one way of diagnosing the crisis that may be useful in eventually helping resolve it. My suggestion is that the problem is that we don't know what quantum gravity is. Or, put another way, the reason we haven't found a theory is because we don't know what we are looking for.

So, what is quantum gravity? Quantum gravity is whatever satisfies the constraints, or criteria of acceptance, we take to define quantum gravity.

A constraint is a principle, or feature, that physicists assume the new theory must satisfy. Physicists impose these in their search in order to narrow down the space of possible theories so that they're not left groping around completely in the dark.

So, what are the constraints on quantum gravity?

Constraining the Search

The constraints on quantum gravity are both empirical (coming from observational and experimental data) and non-empirical. The empirical constraints usually take

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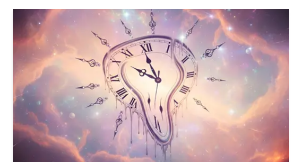
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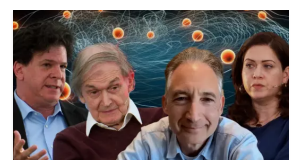
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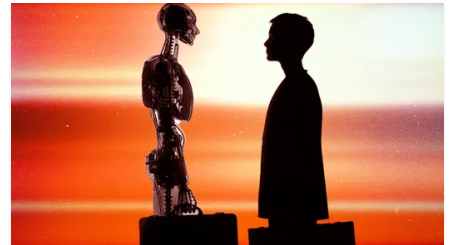
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